

Lab Report

CSE 210 – Digital Logic Design

Student Name: Mohammed Nasser Almarhoon
Student ID: 2240005893

Report No: 1
Date: 22/1/2025

Aim: Study the basic logic gates and verify the outputs using truth tables.

Task#1: Verification of AND Gate

Table 1.1 (AND GATE)			
Inputs		Outputs	
		Theory	Observation
A	B		
0	0	0	0
0	1	0	0
1	0	0	0
1	1	1	1

Task #2 Verification of OR gate

Table 1.2 (OR GATE)			
Inputs		Outputs	
		Theory	Observation
A	B		
0	0	0	0
0	1	1	1
1	0	1	1
1	1	1	1

Task#3 Verification of NOT gate

Table 1.3 (NOT GATE)		
Inputs	Outputs	
	Theory	Observation
A		
0	1	1
1	0	0

Result: Gates truth tables are successfully verified.

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Student Name: Mohammed Nasser Almarhoon
Student ID: 2240005893

Report No: 2
Date: 29/1/2025

Aim: Study universal logic gates and verify their truth tables.

Task#1: Verification of NAND Gate

Table 2.1 (NAND GATE)			
Inputs		Outputs	
		Theory	Observation
A	B	F	F
0	0	1	1
0	1	1	1
1	0	1	1
1	1	0	0

Task #2 Verification of NOR gate

Table 2.2 (NOR GATE)			
Inputs		Outputs	
		Theory	Observation
A	B	F	F
0	0	1	1
0	1	0	0
1	0	0	0
1	1	0	0

Task#3 Verification of XOR gate

Table 2.3 (XOR GATE)			
Inputs		Outputs	
		Theory	Observation
A	B	F	F
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	0

Result: universal Gates truth tables are successfully verified.

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Student Name: Mohammed Nasser Almarhoon
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Report No: 3
Date: 12/2/2025

Implement the SOP function $F = XY + XZ$ on breadboard and find the minterms.

Task#1 Implement the following function $F = XY + XZ$ on bread board.

Table 3-1

Input			Outputs	
X	Y	Z	Theory	Observation
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	0	0
1	0	0	0	0
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

Task#2 Convert the function $F = XY + XZ$ into POS form and find the maxterms.

$$F = XY + XZ$$

$F = XY + XZ$
 $F = X(Y + Z)$
 $F = (X + YY')(Y + Z + XX')$
 $F = (X + YY')(Y + Z + XX')$
 $F = (X + Y + ZZ')(X + Y' + ZZ')(X + Y + Z)(X' + Y + Z)$
 $F = (X + Y + Z)(X + Y + Z')(X + Y' + Z)(X + Y' + Z')(X + Y + Z)(X' + Y + Z)$
 $F = (X + Y + Z)(X + Y + Z')(X + Y' + Z)(X + Y' + Z')(X' + Y + Z)$
 $F = 000.001.010.011.100$

 $F = M_0, M_1, M_2, M_3, M_4$

Result: Use simplification methods for Boolean functions.

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CSE 210 – Digital Logic Design

Student Name: Mohammed Nasser Almarhoon
Student ID: 2240005893

Report No: 4
Date: 12/2/2025

- Design and implementation XOR gate with NAND gates only.

Task #1 Constructing XOR gate with NAND gates:

Table 4-1

Inputs		Outputs							
		Theory				Observation			
A	B	F1	F2	F3	F4	F1	F2	F3	F4
0	0	1	1	1	0	1	1	1	0
0	1	1	1	0	1	1	1	0	1
1	0	1	0	1	1	1	0	1	1
1	1	0	1	1	0	0	1	1	0

Task#2 Write down the Boolean expression for F1, F2, F3 and F4 in the space provided:

$$F1 = (A * B)'$$

$$F2 = A + B'$$

$$F3 = A' + B$$

$$F4 = (A(+))B$$

Task #3 : Using [falstad simulator](#), simulate the circuit in figure 4-1(b) and verify the results with Table 4-1.

Result: Use simplification methods for Boolean functions

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CSE 210 – Digital Logic Design

Student Name: Mohammed Nasser Almarhoon
Student ID: 2240005893

Report No: 5
Date: 19/2/2025

Design and implementation of combinational circuit.

Task#1 Design a simple combinational circuit for the given scenario:

Table 5-1

Inputs			Outputs	
			Theory	Observation
P	Q	R	S	S
0	0	0	1	1
0	0	1	1	1
0	1	0	1	1
0	1	1	1	1
1	0	0	0	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Task#2 : For the output S, write a simplified Boolean expression using Boolean theorems.

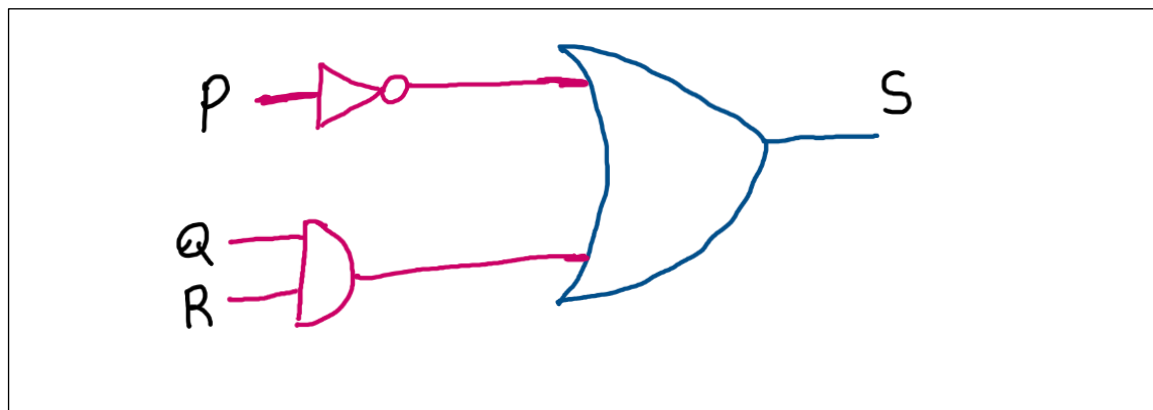
$F = \sum(0, 1, 2, 3, 7)$

$$= \bar{P}\bar{Q}\bar{R} + \bar{P}\bar{Q}R + \bar{P}Q\bar{R} + \bar{P}QR + P\bar{Q}\bar{R}$$
$$= \bar{P}\bar{Q}(\bar{R}+R) + \bar{P}Q(\bar{R}+R) + P\bar{Q}\bar{R}$$
$$= \bar{P}(\bar{Q}+Q) + P\bar{Q}\bar{R}$$

$$= \bar{P} + P\bar{Q}\bar{R}$$
$$= (\bar{P}+P) \cdot (\bar{P}+Q\bar{R})$$
$$= 1 \cdot \bar{P} + Q\bar{R}$$



Task #3 : Draw the logic circuit for simplified Boolean expression.



Result: Design combinational and sequential logic circuits

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Student Name: Mohammed Nasser Almarhoon
Student ID: 2240005893

Report No: 6
Date: 26/2/2025

- Design and implementation of Half Subtractor and Full Subtractor Circuit

Task #1 Constructing HS circuit:

Table 6-1

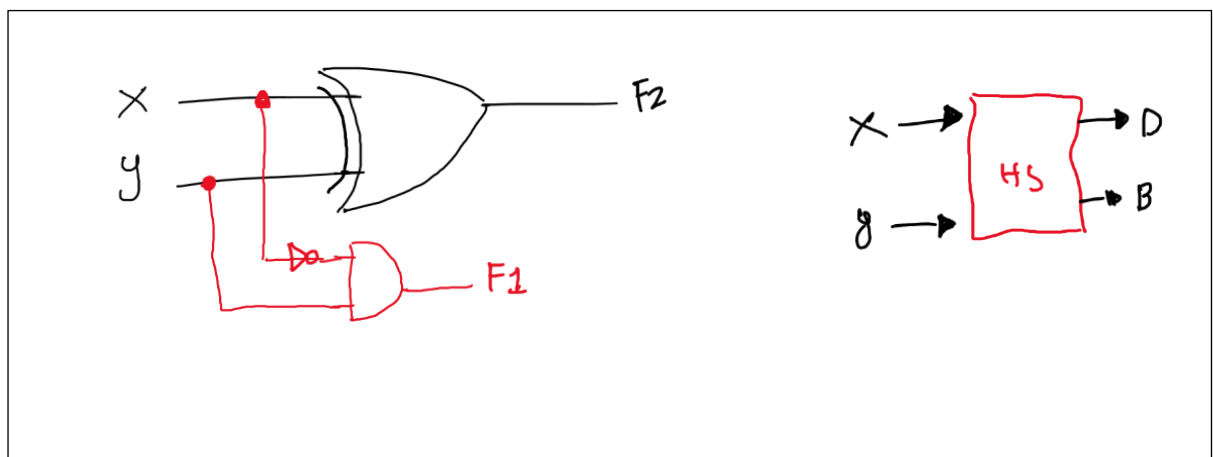
Inputs		Outputs			
		Theory		Observation	
A	B	F1	F2	F1	F2
0	0	0	0	0	0
0	1	1	1	1	1
1	0	0	1	0	1
1	1	0	0	0	0

Task #2 Constructing a Full Subtractor circuit:

Table 6-2

Inputs			Outputs			
			Theory		Observation	
A	B	C	F3	F5	F3	F5
0	0	0	0	0	0	0
0	0	1	1	1	1	1
0	1	0	1	1	1	1
0	1	1	1	0	1	0
1	0	0	0	1	0	1
1	0	1	0	0	0	0
1	1	0	0	0	0	0
1	1	1	1	1	1	1

Task #3 : Draw the HS .



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- **Result:** Simulate combinational and sequential circuits, and predict their behavior .

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Student Name: Mohammed Nasser Almarhoon
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Report No: 7
Date: 5/3/2025

- Design and implementation of Decoder Circuit

Task#1 Constructing a 2-to-4 line Decoder with Basic Gates:

Table 7-1

Inputs		Outputs							
		Theory				Observation			
A	B	F1	F2	F3	F4	F1	F2	F3	F4
0	0	1	0	0	0	1	0	0	0
0	1	0	0	1	0	0	0	1	0
1	0	0	1	0	0	0	1	0	0
1	1	0	0	0	1	0	0	0	1

Task#2 Constructing a 4-to-10 Decoder with TTL IC:

Table 7-2

	Inputs				Outputs																			
					Theory										Observation									
	D	C	B	A	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1
1	0	0	0	1	1	0	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1
2	0	0	1	0	1	1	0	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1
3	0	0	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1
4	0	1	0	0	1	1	1	1	0	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
5	0	1	0	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	0	1	1	1	1
6	0	1	1	0	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	0	1	1	1
7	0	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	0	1	1
8	1	0	0	0	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	0	1
9	1	0	0	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	0

- Result:** Design combinational and sequential logic circuits.

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Student Name: Mohammed Nasser Almarhoon
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Report No: 8
Date: 9/4/2025

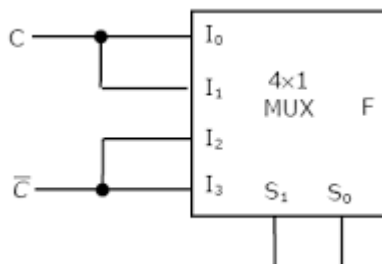
- **Aim:** Implement a function using a Multiplexer

Task#1 Implement $F(C, B, A) = \sum(1, 2, 6, 7)$ using a 8x1 MUX:

Table 8.1:

Input			Output	
A	B	C	Theory	Observation
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	0
1	0	0	0	0
1	0	1	0	0
1	1	0	1	1
1	1	1	1	1

Task#2 Constructing a 4-to-10 Decoder with TTL IC:



A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

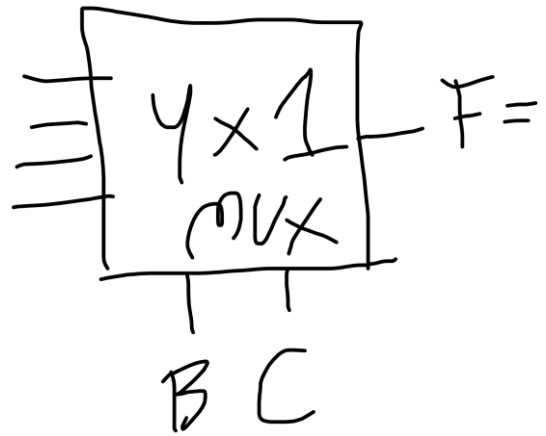
$$F = \sum(1,3,4,6)$$

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Task#3 Find the minterms for the given multiplexes gates :

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

$$F = \sum (1,3,5,6)$$



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Student Name: Mohammed Nasser Almarhoon
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Report No: 9
Date: 16/4/2025

- **Aim:** Practice to learn about the D and JK flip-flop operations

Task#1 Constructing a J-K Flip- Flop with R-S flip flop:

Table 9.1

CK	Input	Output	
	D (A1)	Theory (Q_{n+1})	Observation (F6)
1	1	1	1
0	0	1	1
1	0	0	0
1	1	1	1
0	0	1	1
0	1	1	1

Task#2 Constructing a J-K Flip- Flop with R-S flip flop:

Table 9.2:

CK	Input		Output	
	J	K	Theory (Q_{n+1})	Observation (F6)
1	1	0	1	1
0	0	1	1	1
0	0	0	1	1
1	0	1	0	0
1	0	0	0	0
1	0	1	0	0
1	1	0	1	1
1	1	1	0	0

- **Objective :** Practice to learn about the D and JK flip-flop operations

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Student Name: Mohammed Nasser Almarhoon
Student ID: 2240005893

Report No: 10
Date: 3/5/2025

Aim: Design a Sequential circuit using D flip-flops for a system which has the following state diagram.

Task #1: Setting up the D flip-flop circuit

Table 10-1

State Table

Present State	Input		Next State
<u>A</u>	<u>X</u>	<u>Y</u>	<u>A(t+1)</u>
<u>0</u>	<u>0</u>	<u>0</u>	<u>0</u>
<u>0</u>	<u>0</u>	<u>1</u>	<u>1</u>
<u>0</u>	<u>1</u>	<u>0</u>	<u>1</u>
<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>
<u>1</u>	<u>0</u>	<u>0</u>	<u>1</u>
<u>1</u>	<u>0</u>	<u>1</u>	<u>0</u>
<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>
<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>

☐ Choose the type of flip-flop to be used:

D flip flop

☐ How many flip flops are required?

One filf flop

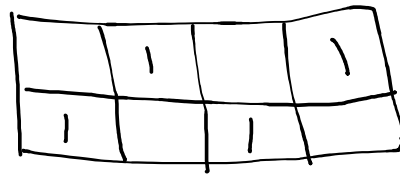
☐ Flip-Flop Input & output equations:

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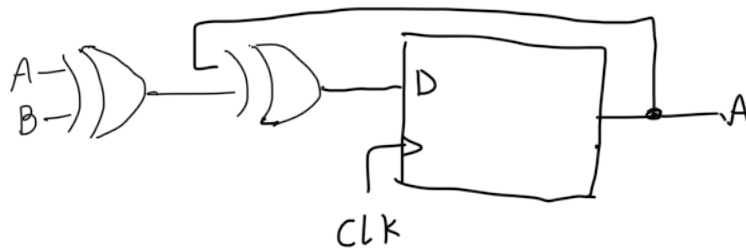
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$$D_A = A \oplus x \oplus y$$

$$A(t+1) = A \oplus x \oplus y$$



- ☐ Draw the sequential circuit with D flip flop:



Task #2:

Simulate the designed sequential circuit (from previous step) using falstad simulator [<http://www.falstad.com/circuit/>] and verify the state table 10-1.

Task #3:

Determine the **next state** and **output** for starting state a (0), and the input sequence **01, 10, 00, 10, 01** for the figure 10.1.

Table 10-2

Present State	Input		Next State	Output
0	0	1	<u>1</u>	<u>1</u>
1	1	0	<u>0</u>	<u>0</u>
0	0	0	<u>0</u>	<u>0</u>
0	1	0	<u>1</u>	<u>1</u>
1	0	1	<u>0</u>	<u>0</u>

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- **Objective :** Practice to learn about the D and JK flip-flop operations

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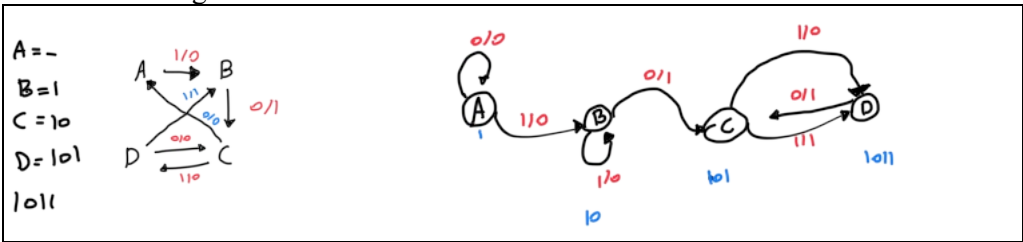
Student Name: Mohammed Nasser Almarhoon
Student ID: 2240005893

Report No: 11
Date: 3/5/2025

Aim: Design combinational and sequential logic circuits

Task#1:

- ☐ Draw a state diagram



- ☐ Table 11-1

State Table

Present State		Input	Next State		Flip Flop Inputs				Output
<u>A</u>	<u>B</u>	<u>X</u>	<u>A(t+1)</u>	<u>B(t+1)</u>	<u>Ja</u>	<u>Ka</u>	<u>Jb</u>	<u>Kb</u>	<u>Y</u>
0	0	0	0	0	0	x	0	x	0
0	0	1	0	1	0	x	1	x	0
0	1	0	1	0	1	x	x	1	0
0	1	1	0	1	0	x	x	0	0
1	0	0	0	0	x	1	0	x	0
1	0	1	1	1	x	0	1	x	0
1	1	0	1	0	x	0	x	1	0
1	1	1	0	1	x	1	x	0	1

- ☐ Choose the type of flip-flop to be used:

Jk flipflop

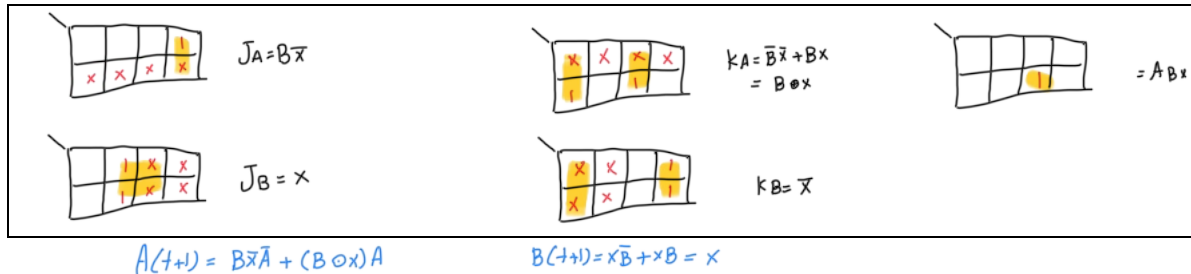
- ☐ How many flip flops are required?

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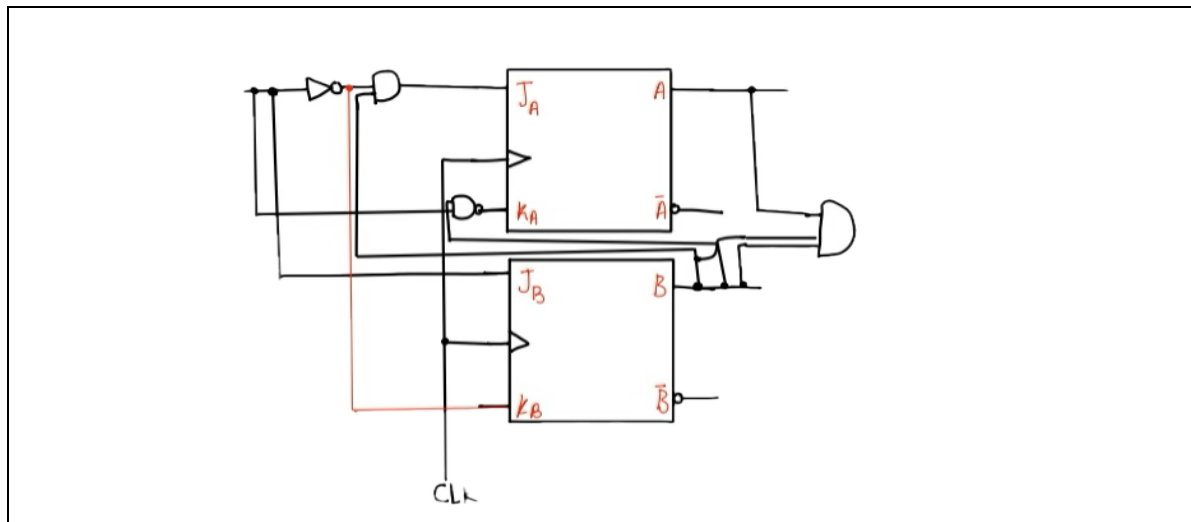
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2 flipflop

- Flip-Flop Input & output equations:



- Draw the sequential circuit with D flip flop:



Task#2:

Simulate the designed sequential circuit (from previous step) using falstad simulator [<http://www.falstad.com/circuit/>] and verify the truth table 11-1.

Task#3:

Draw a **state diagram** of 1011 sequence detector for non-overlap case.



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- **Objective :** Practice to learn about the D and JK flip-flop operations

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Student Name: Mohammed Nasser Almarhoon
Student ID: 2240005893

Report No: 12
Date: 3/5/2025

Aim: Design combinational and sequential logic circuits

Task#1: Design combinational and sequential logic circuits.

Clear	FF0	FF1	FF2	FF3
1001	0	0	0	0
0100	1	0	0	0
0010	0	1	0	0
0001	0	0	1	0
0000	0	0	0	1

Task#2:

Try your solution in the simulator [<http://www.falstad.com/circuit/>] for input sequence **1010**, until you find the correct design and fill the truth table 12-2.

Table 12-1

Clear	FF0	FF1	FF2	FF3
1010	0	0	0	0
0101	0	0	0	0
0010	1	0	0	0
0001	0	1	0	0
0000	1	0	1	0

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- **Objective :** Practice to learn about the D and JK flip-flop operations